# Orbital Curbside

## A garbage truck for space junk.

One spacecraft, ten pickups per trip. Earth does the steering.



Every dot is a real tracked object, from the 16 Oct 2026 public catalogue.

# Orbital Curbside

Cleaning up space junk ten pieces at a time, priced like a garbage route.

[headshot]

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# Space junk is multiplying, and removing one piece costs $40–93 million. We can do it for about $12 million.

EVERYONE ASKS

## How do we grab a piece of junk?

WE ASKED

## Why does every piece need its own rocket?

**PROBLEM**

Dead satellites and rocket stages pile up. Every crash makes thousands more pieces.

**OUR FIX**

One spacecraft collects **10 pieces per trip**, on a route, like a garbage truck.

**WHO PAYS**

Space agencies first, then satellite operators and insurers. A fixed fee per piece.

**PRICE PER PIECE**

**$12.2M**

our model, launch included

Price: Orbital Curbside cost model, base assumptions (appendix). Today's prices: slide 7.

# 47,000 objects circle Earth at 27,000 km/h. Two in three are junk.



**16,000**
still working

**31,000**
dead satellites, old rocket stages and crash fragments

WHY IT MATTERS TO YOU

Maps, weather forecasts and the internet run on satellites flying through this. The Space Station has swerved to dodge junk **41 times.**

ESA DISCOSweb statistics (Oct 2026) · CelesTrak catalogue, 16 Oct 2026 · NASA ODQN 29(3)

# When two big pieces collide, they become thousands. The big ones are the bombs.

**47,000** tracked objects. Over half are small crash fragments: too many and too small to collect one by one.

**~500** **big dead rocket stages and satellites,** 1–9 tonnes each. One crash in 2009 made 2,200 new pieces.

**50** **the worst of them.** Removing just these would cut the risk of debris-making crashes **about in half.**

HOW MANY REMOVALS ARE ENOUGH?

**~5 a year**

NASA models: removing about 5 big objects a year keeps low orbit from getting worse.

**0 so far**

big dead objects removed by a cleanup spacecraft so far. The first capture attempt (Japan, ADRAS-J2) is planned for 2027–28.

ESA Space Environment Report 2026 · McKnight, IAA memo; Top 50 update, IAC 2025 · CelesTrak, Iridium 33 / Cosmos 2251 (2009) · Liou & Johnson, NASA (2009)

# Each cleanup today costs $40–93 million per piece, because each one buys its own rocket.

| WHO PAID | PIECES | PRICE PER PIECE |
|---|---|---|
| Europe (ESA), 2020 | 1 | $93M |
| Japan (JAXA), 2024 | 1 | $82M |
| UK Space Agency, 2026 | 2 | $40M |
| US Space Force (SDA), 2026 | 1 | $52.5M |

**The root cause isn't the grabbing. It's the business model.** It's like hiring a whole garbage truck for every trash bag.

WHY NOW

## 5 years

New US rule: dead satellites must come down within 5 years.

## 1 in 10

may fail to get down on their own. Someone has to pick them up.

## $1.1–1.6B

analysts' estimate of the debris-removal market, 2025.

ESA ClearSpace-1 €86M (2020) · JAXA ADRAS-J2 ≈¥13.2B (2024) · UK ADR >£60M for 2 (Oct 2026) · Starfish/SDA $52.5M incl. options (Jan 2026) · FCC 22-74, FCC 26-47 · Mordor Intelligence; ResearchAndMarkets

# Three customers already have a reason to pay. Agencies pay first.

## 1

### Space agencies

ESA · Japan · UK · US

**WHY THEY CARE**

Their old rockets are up there, and they already sign cleanup contracts.

**HOW THEY BUY**

Contracts paid at milestones.

## 2

### Satellite companies

Iridium · OneWeb · Planet

**WHY THEY CARE**

Must bring dead satellites down within 5 years. Some will fail.

**HOW THEY BUY**

A fixed fee per satellite removed.

## 3

### Insurers

Space underwriters

**WHY THEY CARE**

They pay out when satellites are lost or hit.

**HOW THEY BUY**

Removal bundled with policies, once it's proven.

“I expect the cost is really in the millions or tens of millions, at which price I know it doesn't make sense.”

Matt Desch, CEO of Iridium (2019), on removing its dead satellites. **Satellite companies need a much lower price. A route gives it.**

Numbered by who pays first · Quote: C. Henry, SpaceNews, 30 Dec 2019

# First route: 10 old rocket stages from the US, India and France.

**4**
India (PSLV)

**4**
France (Ariane)

**2**
USA (Agena, Delta)

### Where the cheap rockets go

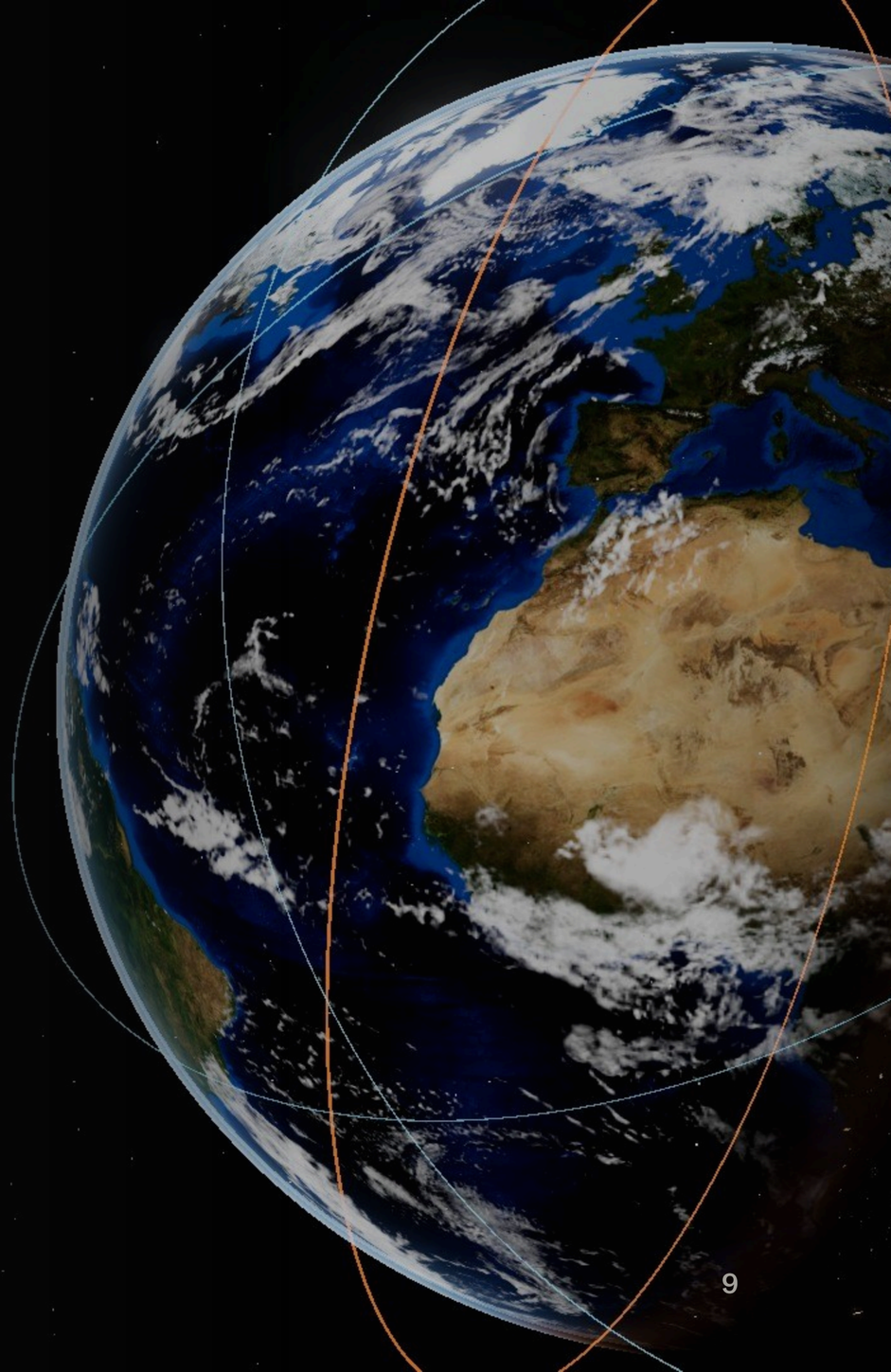
Shared SpaceX launches drop satellites right in their orbit band. First pickup: 80 days after launch.

### No permission problem

Every object belongs to the country that launched it. These owners' agencies already pay for cleanup.

### The bigger prize, later: 142 Soviet-era SL-8 stages

A type on experts' list of the 50 most dangerous objects. Russia must agree; about $13M each.



Our analysis of the CelesTrak catalogue, 16 Oct 2026 · Outer Space Treaty (1967), Art. VIII

# Today, every piece gets its own rocket. We run a route: one trip, ten pickups.

TODAY

**One rocket per piece**

10 pieces = 10 launches



**$40–93M**

per piece

ORBITAL CURBSIDE

**One truck, ten stops**

10 pieces = 1 launch



**$12.2M**

per piece, our model

**Our promise:** space cleanup priced like a garbage route. A fixed fee per piece, paid only when it's confirmed coming down.

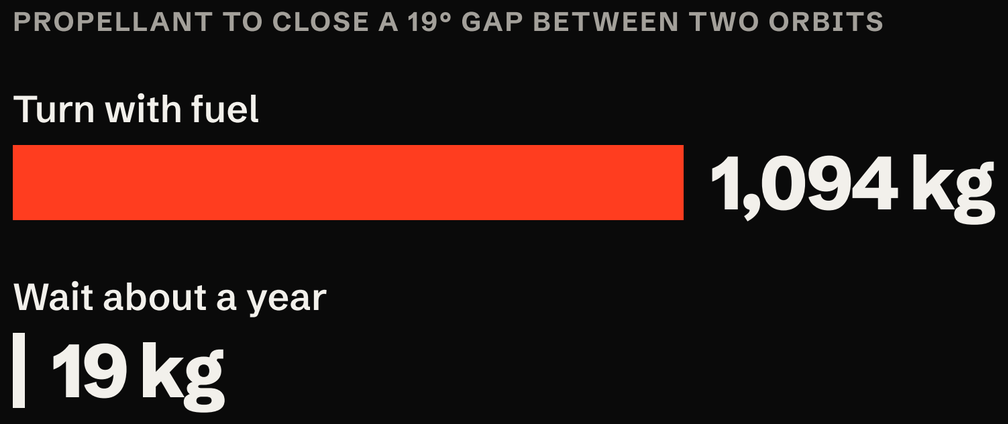
Our figure: model estimate, launch included, base assumptions (appendix) · Today's prices: slide 7

# We don't chase the junk. We let Earth bring it to us.

**Every orbit slowly swings around Earth,**
like the hand of a clock. Lower orbits swing faster, because Earth bulges at the equator.

**So the truck parks a little higher or lower**
than its next target, and waits until the two line up, like clock hands meeting.

PROPELLANT TO CLOSE A 19° GAP BETWEEN TWO ORBITS




**57× less.** Our planner trades waiting time against fuel to find the cheapest route.

Our model · J2 nodal precession (Vallado, 2013)

# Four steps, ten times per trip.



1

## Ride along

The truck rides a shared SpaceX rocket, the cheapest seat to space.



2

## Drift into line

It climbs to the junk's height and waits for Earth to line up the next target.



3

## Grab and tag

A robot arm holds the old rocket. The truck latches on a small rocket kit.



4

## Send it down

The kit fires. The old rocket falls and burns up. The truck drives on.

Renders from our 3D concept model, sized by our math. Not a flight design.

# About as heavy as a car.

**Electric engines**
Run on xenon gas; about 5× more fuel-efficient than chemical rockets.

**Robot arm**
Holds each old rocket steady while a kit is attached.

**10 rocket kits**
One per pickup. Each sends one piece down to burn up.

**Solar wings**
Turn sunlight into power for the engines.

WEIGHT AT LAUNCH

**1,650 kg**



**852 kg** 10 kits

**600 kg** truck

**198 kg** xenon

Concept sized from our model (first route). Not a flight design. Thruster class: Busek BHT-1500.

# Is a high-schooler's math right? We checked it against NASA JPL.

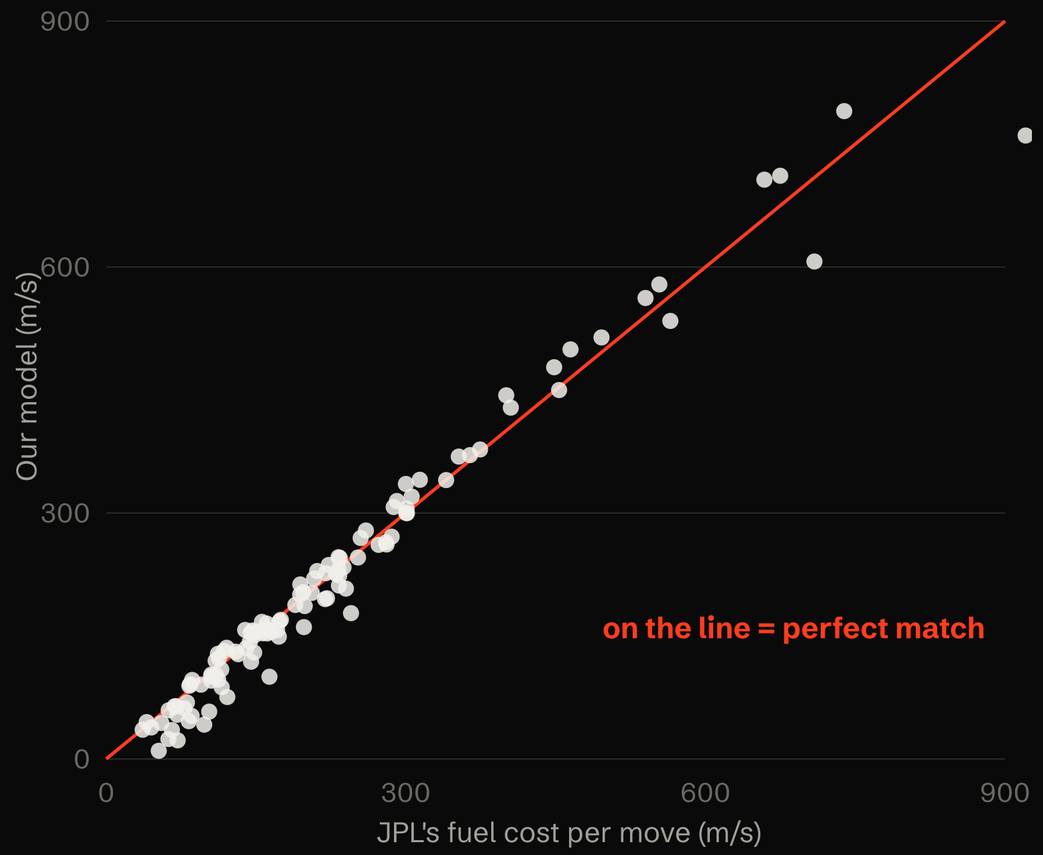
In 2017 the European Space Agency ran a contest to plan routes exactly like ours. NASA's Jet Propulsion Laboratory won, and published every move.

**2.2%**

our fuel totals vs JPL's, across all 113 moves

**0.06°**

our orbit predictions vs SGP4, the model used to track real satellites




Each dot is one of the 113 moves in JPL's winning plan, recomputed by our model.

Petropoulos et al. (2018), GTOC9 JPL results · Vallado et al. (2006), SGP4 · 0.30° on our first route

# Grab it. Tag it. Send it down.

**10**
old stages removed per trip

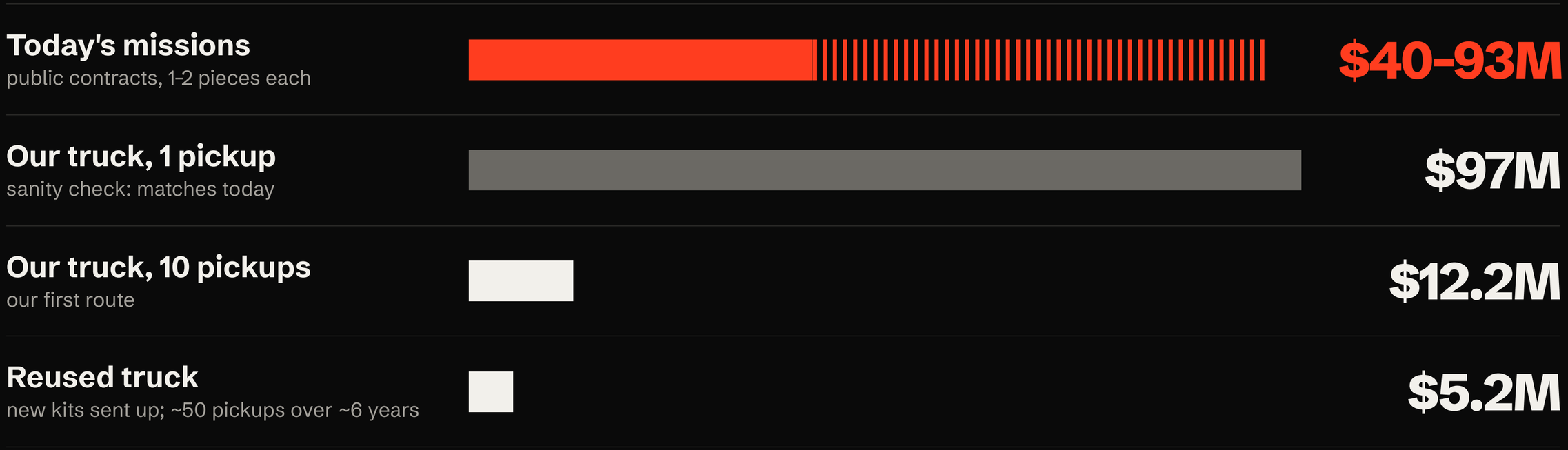
**490 days**
launch to last stop (16 months)

**198 kg**
of xenon fuel for the whole trip



3D concept model, sized from our mass budget · route: our planner + SGP4. Not a flight design.

# More pickups per trip, lower price per piece.




Cost per piece removed, launch included. **80% of 20,000 simulated cases land between $11.2M and $15.4M.** Biggest unknown: what the truck itself costs ($50–115M).

Our model; inputs from NASA OTPS (2023), SpaceX rideshare prices, Busek · assumptions in appendix

# What could stop us, and what we do about it.

RISK

## Grabbing it

Nobody has grabbed a tumbling old rocket yet. **We partner with whoever proves it** (Japan tries in 2027–28).

RISK

## Cost

Truck and kit prices are estimates. **I'm testing them with experts.**

RISK

## Permission

Every object belongs to its country. **First route: US, Indian and French stages.**

RISK

## Falling pieces

Heavy objects are **steered to burn up over the open ocean.**

**2026–27 · now**

Talk to customers and experts; sharpen the model

**2027–28**

Sign a partner for the arm and spacecraft

**2028–29**

Small test trip: 2–3 pickups

**2029+**

First 10-pickup truck

Outer Space Treaty (1967), Art. VIII · JAXA/Astroscale ADRAS-J2 · dates after 2027 are targets

# Every crash makes more junk. Let's take out the space trash before it takes out what we rely on.

1. We don't need to remove it all: **taking out the worst 50 cuts the danger about in half.**
2. Cleanup costs $40–93M per piece today. **Our route: about $12M, launch included.**
3. Earth does the steering, and **the math is checked against NASA JPL.**

THE ASK

**[$__] in pre-seed funding for 12 months:** finish customer interviews, design the rocket kit, sign a test-trip partner.

Scan to fly the route

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Earth imagery: NASA Visible Earth / Earth Observatory (Blue Marble, Black Marble, clouds). Renders: our own.

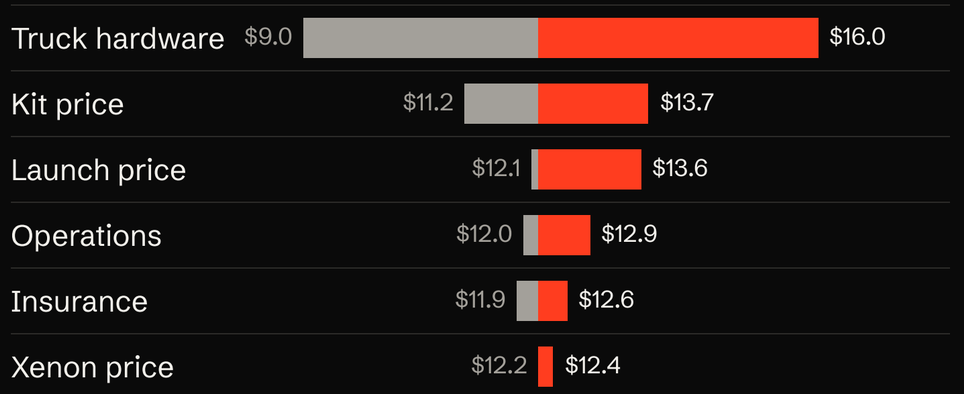
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Full list with links: written paper, references [1]–[28]

# Every assumption, and what moves the price.

| INPUT | LOW · BASE · HIGH | SOURCE |
|---|---|---|
| **Truck hardware** | $50 · 80 · 115M | NASA OTPS (2023); base = assumption |
| **Rocket kit, each** | $0.5 · 1.5 · 3.0M | Assumption: no public price |
| **Launch** | $7.0 · 7.5 · 15k/kg | SpaceX rideshare to 550 km; truck climbs |
| **Xenon** | $2.8 · 2.8 · 12k/kg | Market reports (2024-26) |
| **Operations** | $3 · 5 · 10M/yr | Assumption |
| **Insurance** | 5 · 8 · 12% | Assumption, of hardware + launch |
| **Fuel margin** | 10 · 10 · 25% | Our JPL and SGP4 checks |
| **Truck** | 600 kg dry; Hall thrusters, Isp 1,710 s; 10 days per stop | |
| **Kits** | Solid motor, Isp 285 s; lowers orbit to 250 km; ≈85 kg each (first route) | |

COST PER PIECE IF ONE INPUT GOES LOW → HIGH (BASE $12.2M)




**Truck cost matters most.** Fuel barely matters.

Base case: first route, 10 pickups, 490 days from launch, launch included. Sources in paper.